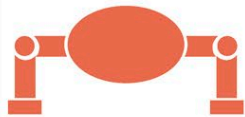


WALKING UPRIGHT

Fossil studies have revealed that one of the main features that distinguishes dinosaurs from other reptiles is their stance. Unlike most reptiles, whose limbs are spread out to the side, dinosaurs were able to walk upright with their limbs underneath their body.



Lizard

With their legs spread out, lizards have a sprawling stance that only allows short bursts of movement.



Crocodile

Crocodylians have flexible ankles that allow them to lift their bodies off the ground if they want to travel quickly on land.



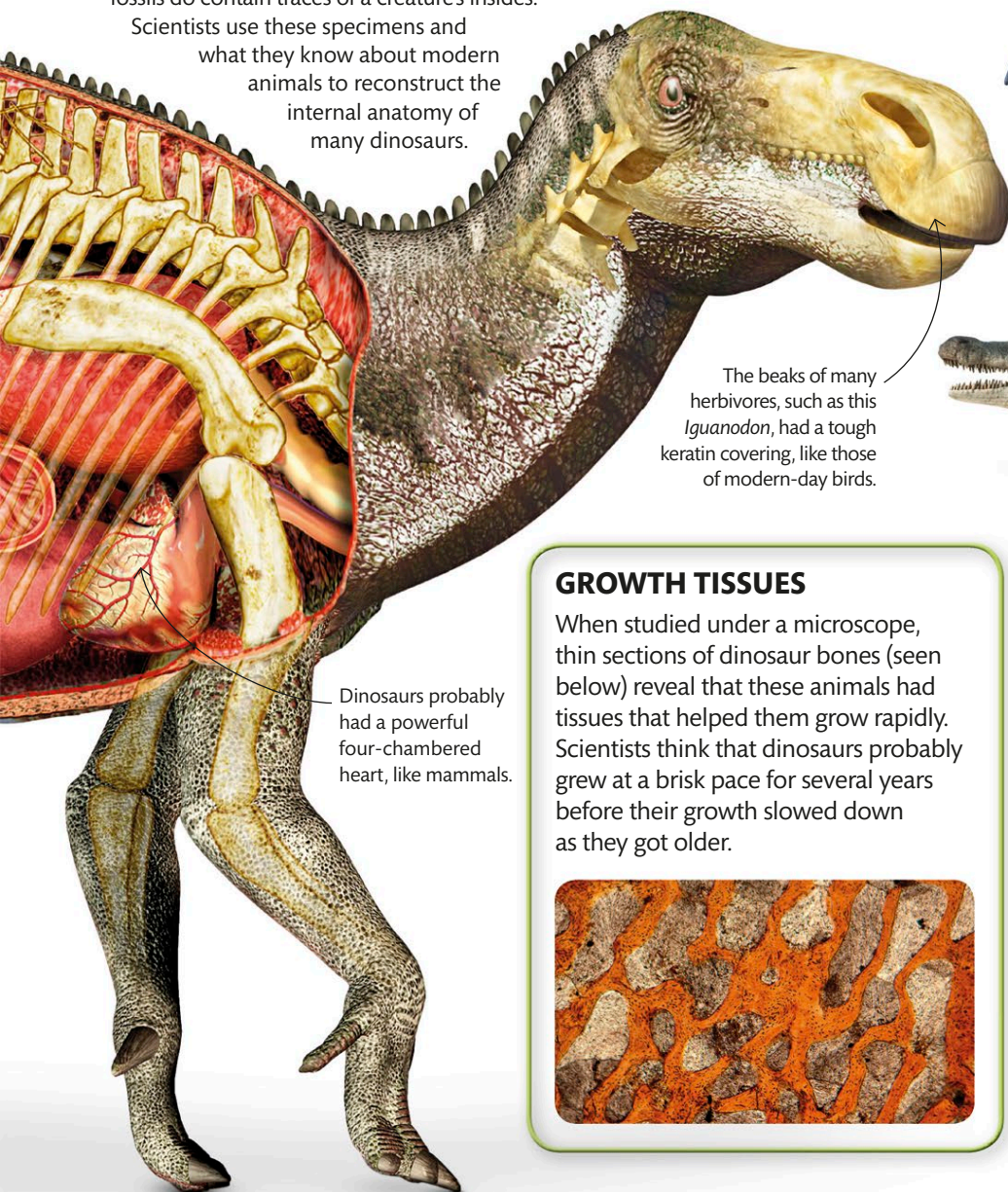
Dinosaur

With their limbs tucked under their bodies, dinosaurs could stand upright and walk efficiently.

Inside a dinosaur

Although the internal organs of dinosaurs were often eaten by scavengers during decomposition or destroyed during fossilization, some exquisitely preserved fossils do contain traces of a creature's insides.

Scientists use these specimens and what they know about modern animals to reconstruct the internal anatomy of many dinosaurs.



The beaks of many herbivores, such as this *Iguanodon*, had a tough keratin covering, like those of modern-day birds.

Dinosaurs probably had a powerful four-chambered heart, like mammals.

GROWTH TISSUES

When studied under a microscope, thin sections of dinosaur bones (seen below) reveal that these animals had tissues that helped them grow rapidly. Scientists think that dinosaurs probably grew at a brisk pace for several years before their growth slowed down as they got older.



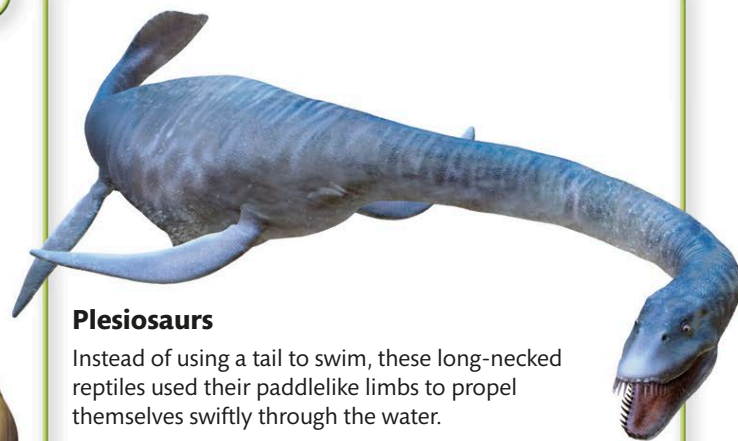
NOT A DINOSAUR

The Mesozoic Era saw a wide range of ancient reptiles flourish on land, in the air, and at sea. However, not all of these creatures were dinosaurs. Some, such as the pterosaurs, were cousins of the dinosaurs, but others were only distantly related to them.



Ichthyosaurs

These large, fishlike creatures were actually marine reptiles that gave birth to live young, just as modern-day mammals do.



Plesiosaurs

Instead of using a tail to swim, these long-necked reptiles used their paddlelike limbs to propel themselves swiftly through the water.



Crocodylomorphs

Relatives of modern crocodylians, creatures in this group evolved a range of different lifestyles and anatomies, with some reaching more than 33 ft (10 m) in length.



Pterosaurs

The first vertebrates to take to the skies, the pterosaurs had an elongated fourth finger, which supported a complex wing membrane.